

U.S. SEMICONDUCTOR: TRADE & LABOR ANALYSIS

Global Semiconductor Demand Study — SIA & OECD Sources | 2025

Overview

This document consolidates accurate global semiconductor demand data from official sources — the Semiconductor Industry Association (SIA) and the Organisation for Economic Co-operation and Development (OECD). Since no structured datasets were available for direct download, data was gathered through manual extraction from published reports and official websites, complemented by AI-assisted structuring and human verification. The result is a transparent, reliable dataset that captures both quantitative sales trends and qualitative strategic context.

Approach Taken

1. Manual Extraction from Official Sources

Monthly global semiconductor sales data for 2025 was gathered directly from SIA market reports, including values in billions of USD, year-over-year changes, and month-over-month variations. Each entry was linked to its official source for full auditability.

2. Contextual Enrichment from OECD

Qualitative insights were extracted from OECD publications, focusing on value-added distribution, demand sectors, and systemic risks. This provided the strategic backdrop to the raw sales figures — connecting market performance to structural vulnerabilities.

3. AI + Human Synthesis

Because no downloadable datasets existed, AI was used to structure the extracted information into clean tables and narratives, while accuracy was manually verified against the original sources. This hybrid approach ensured both efficiency and reliability.

4. Documentation of Strategic Context

OECD warnings about geographic concentration, resilience risks, and diversification attempts were incorporated. This contextual layer elevates the dataset from raw numbers to strategic intelligence.

Data Sources

Institution	Resource	Type of Data
SIA (Semiconductor Industry Association)	Market Data Reports	Monthly global semiconductor sales (USD billions), YoY and MoM changes
OECD	Semiconductors Publication	Value-added distribution, demand sectors, geographic concentration, systemic risk analysis

Key Insights — Strategic Context

Geographic Concentration

75% of semiconductor value added is concentrated in five countries: USA, South Korea, Taiwan, Japan, and China. This creates systemic vulnerability across the entire global supply chain.

Geopolitical Risks

Active risks include U.S.–China trade tensions and potential disruption scenarios involving Taiwan — currently the world's most critical node for advanced chip fabrication.

Limited Supply Chain Resilience

The OECD highlights that the degree of geographic concentration leaves the global semiconductor ecosystem highly exposed to regional disruptions, whether geopolitical, natural, or logistical.

Diversification Attempts

Europe and India are pursuing diversification through subsidies and industrial policy (e.g., EU Chips Act, India Semiconductor Mission), but both remain marginal in global value-added share at present.

Key Insights — Policy & Market Implications

Corporate Dependence

Global technology firms — from automotive manufacturers to consumer electronics brands — rely heavily on this concentrated supply chain. A disruption in any of the five dominant nodes creates cascading effects across industries.

Policy Response

Governments are actively pushing reshoring and subsidies to reduce dependence. Key initiatives include the U.S. CHIPS and Science Act and the European Chips Act, both designed to incentivize domestic fabrication capacity.

Market Outlook

Indicator	Value / Projection
Global semiconductor sales projection (2026)	~\$1 trillion
Concentration of value added	75% in 5 countries (USA, Korea, Taiwan, Japan, China)
Dominant supply-side risk	Taiwan fabrication exposure
Key policy levers	U.S. CHIPS Act, EU Chips Act, India Semiconductor Mission
Emerging diversification markets	Europe, India (currently marginal in global share)

Methodology Notes

- No structured datasets available:** SIA and OECD do not provide bulk-downloadable structured datasets for this data. Manual extraction was the only viable collection method.
- AI-assisted structuring:** AI tools were used to convert extracted text and tables into clean, structured formats. All outputs were manually verified against original source pages.

- **Auditability:** Each data point is linked to its original source URL, enabling independent verification and replication.
- **Scope:** This study covers 2025 monthly sales data and the most recent available OECD strategic analysis of the semiconductor industry.

Data Sources

- [Semiconductor Industry Association \(SIA\) — Market Data](#)
- [OECD — Semiconductors](#)